

Dietary Mutagen Exposure and Risk of Pancreatic Cancer



To investigate the association between dietary exposure to food mutagens and risk of pancreatic cancer, we conducted a hospital-based case-control study at the University of Texas M. D. Anderson Cancer Center during June 2002 to May 2006. A total of 626 cases and 530 noncancer controls were frequency matched for race, sex and age (± 5 years). Dietary exposure information was collected via personal interview using a meat preparation questionnaire. A significantly greater portion of the cases than controls showed a preference to well-done pork, bacon, grilled chicken, and pan-fried chicken, but not to hamburger and steak. Cases had a higher daily intake of food mutagens and mutagenicity activity (revertants per gram of daily meat intake) than controls did. The daily intakes of 2-amino-3,4,8-trimethylimidazo[4,5-f]quinoxaline (DiMeIQx) and benzo(a)pyrene (BaP), as well as the mutagenic activity, were significant predictors for pancreatic cancer ($P = 0.008$, 0.031 , and 0.029 , respectively) with adjustment of other confounders. A significant trend of elevated cancer risk with increasing DiMeIQx intake was observed in quintile analysis ($P_{\text{trend}} = 0.024$). A higher intake of dietary mutagens (those in the two top quintiles) was associated with a 2-fold increased risk of pancreatic cancer among those without a family history of cancer but not among those with a family history of cancer. A possible synergistic effect of dietary mutagen exposure and smoking was observed among individuals with the highest level of exposure (top 10%) to PhIP and BaP, $P_{\text{interaction}} = 0.09$ and 0.099 , respectively. These data support the hypothesis that dietary mutagen exposure alone and in interaction with other factors contribute to the development of pancreatic cancer.